

THE UNIVERSITY OF HONG KONG
FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE

COMP3314/CSIS0314 Machine Learning

Date : May 18, 2015

Time : 9:30 am - 12:30 pm

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DIRECTION TO CANDIDATES

1. Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is the candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Brand and model of the calculator: _____

2. Do not remove supplementary sheets, used or unused, from the Examination Room.
3. Answer ALL questions. Total mark is 100.

For Examiner's use only:									
Q.No.	1	2	3	4	5	6	7	8	Total
Mark									

1. (12 marks) True or False Questions (Give a short explanation for each answer).:

- (a) The training error for a nearest neighbour (1-NN) classifier is always 0.
- (b) The loss function for logistic regression contains multiple local minima.
- (c) Multilayer perceptrons (MLPs) can only represent nonlinear decision boundaries if they contain nonlinear hidden-unit activation functions.
- (d) Making predictions using parameters obtained from MAP estimation accounts for our uncertainty in the learned parameters.
- (e) PCA and MoG models are not useful for classification as they are unsupervised learning models.
- (f) The Naive Bayes assumption corresponds to constraining the covariance of a Gaussian distribution to be diagonal. This means it can only be applied to problems with continuous input data.

2. (16 marks) Short Answer Questions:

- (a) (3 marks) Discuss differences between generative and discriminative approaches to classification. (Hint: Consider the probability distributions modeled by each approach.)

- (b) (2 marks) Name two properties of the clustering model that are learned by the mixture-of-Gaussians model but are not captured by the K-means model.

(c) (3 marks) Consider learning the soft margin SVM

$$\arg \min_{\mathbf{w}, b, \xi} \left\{ C \sum_{n=1}^N \xi_n + \frac{1}{2} \|\mathbf{w}\|^2 \right\} \text{ subject to}$$
$$t_n(\mathbf{w}^T \mathbf{x}_n + b) \geq 1 - \xi_n, \xi_n \geq 0, \forall n.$$

What's the behavior of the margin $\frac{1}{\|\mathbf{w}\|}$ as $C \rightarrow 0$? Explain.

(d) (2 marks) Give two examples of ensemble classifiers.

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(e) (3 marks) Describe the role of gating functions in a mixture of Experts model. What's the consequence of using a *softmax* gating function on the behavior of the classifier?

(f) (3 marks) Explain the exploitation vs. exploration trade-off in reinforcement learning.

3. (10 marks) Logistic Regression

Consider an alternative formulation of logistic regression. Given a training set $\mathcal{D}$ with input $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ and output $\{t_1, \dots, t_N\}$, where $\mathbf{x}_i \in \mathbb{R}^M$, $t_i \in \{-1, 1\}$. We will write the probability of observing $t_i = 1$ given $\mathbf{x}_i$ as $\sigma(y_i)$ (i.e., $P(t_i = 1|\mathbf{x}_i) = \sigma(y_i)$), where $\sigma(a) = (1 + \exp(-a))^{-1}$, $y_i = \mathbf{w}^T \mathbf{x}_i + b$.

Assume the training data is IID (independent and identically distributed), show that the negative log-likelihood can be written as $\sum_{n=1}^N \log(1 + \exp(-y_i t_i))$.

4. (12 marks) Decision Trees

We will build a decision tree to classify whether a student will attend today's lecture or not using the data set below. Recall that the entropy of a collection of training examples of K classes is defined as

$$I = - \sum_{k=1}^K p_k \log_2 p_k,$$

where p_k is the proportion of examples in class k . Note that $0 \log_2 0 \equiv 0$.

Air Quality	Temperature	Humidity	Wind	Attend Lecture
Average	Warm	Low	No	Yes
Poor	Warm	Medium	Yes	Yes
Good	Hot	Medium	No	Yes
Good	Cold	High	Yes	Yes
Poor	Cold	Low	Yes	No
Poor	Warm	High	No	No

(a) (2 marks) What's the entropy of the training examples with respect to the target label "Attend Lecture"?

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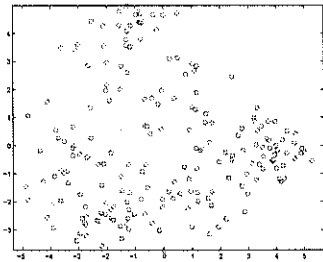
(b) (5 marks) What are the information gains for each of the 4 attributes? Which attribute should be used as the root node of the decision tree?

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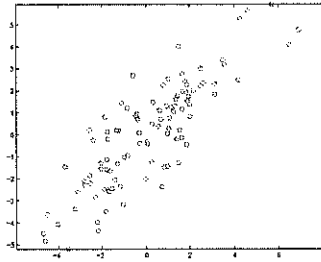
(c) (5 marks) Draw the entire decision tree using the learning algorithm described in class. Show your steps.

5. (12 marks) PCA

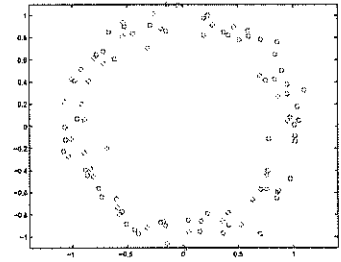
(a) (5 marks) Consider the projection of a high-dimensional data set into a 2-dimensional space using standard PCA. Which of the following result(s) is(are) impossible? Explain.



I



II



III

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(b) (4 marks) PCA relies on the eigendecomposition of the sample covariance matrix $S = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$. What are the contents of matrices $\mathbf{U}$ and $\mathbf{\Lambda}$? Explain in terms of the variance of the data distribution.

(c) (3 marks) Consider applying PCA to a 10 dimensional input data set, suppose we obtained the following sorted eigenvalues after the eigendecomposition step $\{1500, 200, 15, 10, 2, 0.5, 0.1, 0.01, 0.01, 0.01\}$. How many components would you need to keep in order to retain 90% of the total data variance?

6. (15 marks) Mixture of Gaussians

Consider a mixture-of-Gaussians (MoG) model with K components, a data vector $\mathbf{x} \in \mathbb{R}^D$ and a corresponding latent indicator variable $\mathbf{z} \in \{0, 1\}^K$, $\sum_{k=1}^K z_k = 1$.

(a) (2 marks) Write $p(\mathbf{x})$ in terms of $P(\mathbf{z})$ and $p(\mathbf{x}|\mathbf{z})$ using rules of probability.

(b) (5 marks) Define the distributions:

$$P(\mathbf{z}) = \prod_{k=1}^K \pi_k^{z_k}$$

$$p(\mathbf{x}|\mathbf{z}) = \prod_{k=1}^K \mathcal{N}(\mathbf{x}|\mu_k, \Sigma_k)^{z_k}.$$

Show that

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}|\mu_k, \Sigma_k).$$

(c) (3 marks) Explain the meaning of $P(z_k = 1|\mathbf{x})$ in the MoG model, then show that

$$P(z_k = 1|\mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)},$$

(d) (5 marks) Derive and simplify the expression for $\log p(\mathbf{x}, \mathbf{z})$. Why might this expression be easier to work with than $\log p(\mathbf{x})$?

7. (13 marks) EM and Free Energy

Consider a model with continuous observed data $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_N]^T$ and corresponding discrete latent variables $\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_N]^T$. Let θ denote parameters of the model. The *incomplete log-likelihood* is given by

$$\mathcal{L}(\theta) = \log p(\mathbf{X}|\theta) = \log \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z}|\theta).$$

Let $q(\mathbf{Z})$ denote a probability distribution over latent variables. The *free energy* of the model is defined by

$$F(q, \theta) = \sum_{\mathbf{Z}} q(\mathbf{Z}) \log p(\mathbf{X}, \mathbf{Z}|\theta) - \sum_{\mathbf{Z}} q(\mathbf{Z}) \log q(\mathbf{Z}).$$

Let $\theta^{(i-1)}$ denote model parameters estimated from the $(i-1)$ -th iteration of the EM algorithm.

(a) (4 marks) In the E-step of the i -th iteration, set $q^{(i)} = p(\mathbf{Z}|\mathbf{X}, \theta^{(i-1)})$. Show that

$$\mathcal{L}(\theta^{(i-1)}) = F(q^{(i)}, \theta^{(i-1)}).$$

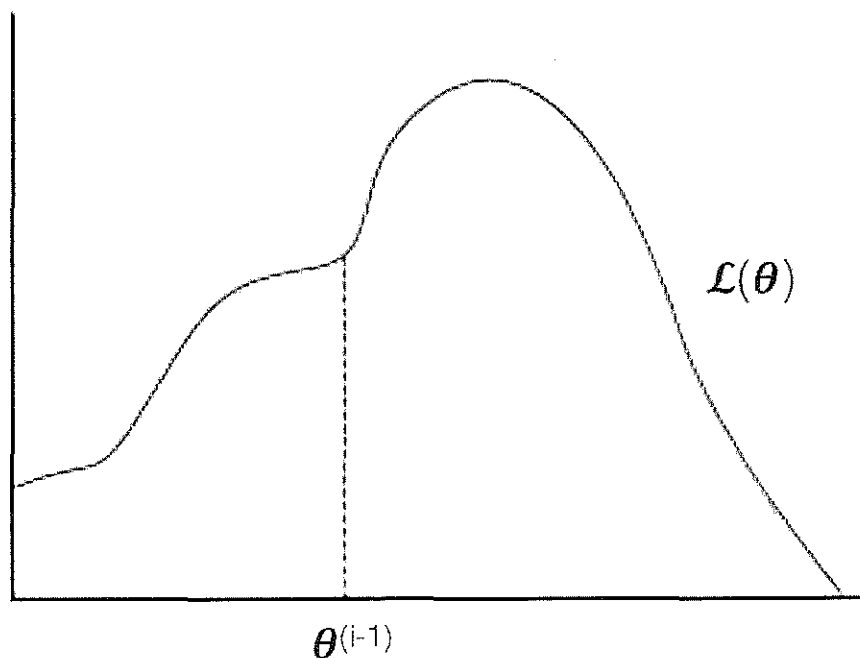
(b) (2 marks) In the M-step, set $\theta^{(i)}$ to maximize the expected complete log-likelihood

$$\theta^{(i)} = \arg \max_{\theta} \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \theta^{(i-1)}) \log p(\mathbf{X}, \mathbf{Z}|\theta).$$

Explain why $\theta^{(i)}$ is also a maximizer of $F(q^{(i)}, \theta)$.

(c) (3 marks) Show that $\mathcal{L}(\theta^{(i)}) \geq \mathcal{L}(\theta^{(i-1)})$ using parts (a) and (b). (Hint: You can assume $\mathcal{L}(\theta) \geq F(q, \theta)$ — free energy is a lower bound on the log likelihood.)

(d) (4 marks) Below is a graph of the log incomplete likelihood $\mathcal{L}(\theta)$ vs. model parameter value (assume θ is 1D). Given the labeled parameter value at iteration $i - 1$, first sketch a plausible curve representing $F(q^{(i)}, \theta)$ as a function of θ , then draw a vertical line at a plausible location for $\theta^{(i)}$.



8. (10 marks) Kernels

- (a) (5 marks) Show that $k(\mathbf{x}, \mathbf{z}) = (\mathbf{x}^T \mathbf{z})^2$ is a valid kernel function by identifying a feature mapping Φ such that $k(\mathbf{x}, \mathbf{z}) = \Phi(\mathbf{x})^T \Phi(\mathbf{z})$, where $\mathbf{x}, \mathbf{z} \in \mathbb{R}^2$.

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(b) (5 marks) The nearest-neighbour (NN) classifier assigns new input vector $\mathbf{x}$ to the same class as the nearest input vector $\mathbf{x}_n$ from the training set, where the distance is defined by $\|\mathbf{x} - \mathbf{x}_n\|^2$. Formulate a generalized version of the NN classifier using the kernel trick.

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